CORE PURE / DIFFERENTIAL EQUATIONS

2nd Order Differential Equations

Worked solutions 12 questions · 118 marks

Questions

Question 1	*6 marks*	2
Question 2	*10 marks*	4
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Question 12	*10 marks*	27

1. **(a)** Find the general solution of the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4y = 0 \qquad \textbf{[2]}$$

(b) Hence, by considering a particular integral of the form $x(A\cos 2x + B\sin 2x)$, find the general solution of the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4y = 8\cos 2x - 12\sin 2x \qquad \textbf{[4]}$$

Solution

(a) For

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4y = 0$$

assume a solution of the form $y = e^{\lambda x}$. Then the auxiliary equation is

$$\lambda^2 + 4 = 0$$

So

$$\lambda = \pm 2i$$

Hence the general solution is

$$y = C_1\cos 2x + C_2\sin 2x$$

(b) From part (a), the complementary solution is

$$y_c = C_1\cos 2x + C_2\sin 2x$$

Since the right-hand side involves $\cos 2x$ and $\sin 2x$, take a particular integral in the given form

$$y_p = x(A\cos 2x + B\sin 2x)$$

Differentiate:

$$\frac{\mathrm{d}y_p}{\mathrm{d}x} = (A\cos 2x + B\sin 2x) + x(-2A\sin 2x + 2B\cos 2x)$$

Differentiate again:

$$\begin{aligned}\frac{\mathrm{d}^2y_p}{\mathrm{d}x^2} &= (-2A\sin 2x + 2B\cos 2x) + (-2A\sin 2x + 2B\cos 2x)\\ &\quad + x(-4A\cos 2x - 4B\sin 2x)\\ &= -4A\sin 2x + 4B\cos 2x - 4x(A\cos 2x + B\sin 2x)\end{aligned}$$

Now substitute into $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4y$:

$$\begin{aligned}\frac{\mathrm{d}^2y_p}{\mathrm{d}x^2} + 4y_p &= [-4A\sin 2x + 4B\cos 2x - 4x(A\cos 2x + B\sin 2x)]\\ &\quad + 4x(A\cos 2x + B\sin 2x)\\ &= 4B\cos 2x - 4A\sin 2x\end{aligned}$$

This must equal the right-hand side:

$$4B\cos 2x - 4A\sin 2x = 8\cos 2x - 12\sin 2x$$

Equating coefficients gives

$$\begin{aligned}4B &= 8\\ -4A &= -12\end{aligned}$$

so

$$B = 2, \qquad A = 3$$

Therefore

$$y_p = x(3\cos 2x + 2\sin 2x)$$

Hence the general solution is

$$y = y_c + y_p = C_1 \cos 2x + C_2 \sin 2x + x(3\cos 2x + 2\sin 2x)$$

2. Find the particular solution of the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 4\frac{\mathrm{d}y}{\mathrm{d}t} + 5y = 8\cos t$$

given that, when $t = 0$, $y = 3$ and $\frac{\mathrm{d}y}{\mathrm{d}t} = 1$. **[10]**

Solution

First solve the associated homogeneous equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 4\frac{\mathrm{d}y}{\mathrm{d}t} + 5y = 0$$

Its auxiliary equation is

$$\lambda^2 + 4\lambda + 5 = 0$$

Using the quadratic formula:

$$\lambda = \frac{-4 \pm \sqrt{4^2 - 4(1)(5)}}{2(1)}$$

$$= -2 \pm i$$

Hence the complementary function is

$$y_c = e^{-2t}(C_1\cos t + C_2 \sin t)$$

Now find a particular integral for

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 4\frac{\mathrm{d}y}{\mathrm{d}t} + 5y = 8\cos t$$

Since the forcing term is $8\cos t$, try

$$y_p = A\cos t + B\sin t$$

Then

$$\frac{\mathrm{d}y_p}{\mathrm{d}t} = -A\sin t + B\cos t$$

and

$$\frac{\mathrm{d}^2y_p}{\mathrm{d}t^2} = -A\cos t - B\sin t$$

Substituting into the differential equation:

$$\begin{aligned}\frac{\mathrm{d}^2y_p}{\mathrm{d}t^2} + 4\frac{\mathrm{d}y_p}{\mathrm{d}t} + 5y_p &= (-A\cos t - B\sin t) + 4(-A\sin t + B\cos t) + 5(A\cos t + B\sin t)\\ &= (-A + 4B + 5A)\cos t + (-B - 4A + 5B)\sin t\\ &= (4A + 4B)\cos t + (-4A + 4B)\sin t\end{aligned}$$

This must equal $8\cos t$, so equating coefficients gives

$$4A + 4B = 8, \qquad -4A + 4B = 0$$

Divide both equations by 4:

$$A + B = 2, \qquad -A + B = 0$$

From $-A + B = 0$, we get

$$B = A$$

Substitute into $A + B = 2$:

$$A + A = 2$$

$$2A = 2$$

$$A = 1$$

Therefore

$$B = 1$$

So the particular integral is

$$y_p = \cos t + \sin t$$

Hence the general solution is

$$y = e^{-2t}(C_1 \cos t + C_2 \sin t) + \cos t + \sin t$$

Now use the initial conditions.
Given $y(0) = 3$, substitute $t = 0$:

$$\begin{aligned} 3 &= e^0(C_1 \cos 0 + C_2 \sin 0) + \cos 0 + \sin 0 \\ &= 1(C_1 \cdot 1 + C_2 \cdot 0) + 1 + 0 \\ &= C_1 + 1 \end{aligned}$$

So

$$C_1 + 1 = 3$$

$$C_1 = 2$$

Next differentiate the general solution:

$$\begin{aligned} \frac{\mathrm{d}y}{\mathrm{d}t} &= \frac{\mathrm{d}}{\mathrm{d}t}\left[e^{-2t}(C_1 \cos t + C_2 \sin t)\right] + \frac{\mathrm{d}}{\mathrm{d}t}(\cos t + \sin t) \\ &= -2e^{-2t}(C_1 \cos t + C_2 \sin t) + e^{-2t}(-C_1 \sin t + C_2 \cos t) - \sin t + \cos t \\ &= e^{-2t}\left[(-2C_1 + C_2)\cos t + (-C_1 - 2C_2)\sin t\right] - \sin t + \cos t \end{aligned}$$

Given $\frac{\mathrm{d}y}{\mathrm{d}t}(0) = 1$, substitute $t = 0$:

$$\begin{aligned} 1 &= e^0\left[(-2C_1 + C_2)\cos 0 + (-C_1 - 2C_2)\sin 0\right] - \sin 0 + \cos 0 \\ &= 1\left[(-2C_1 + C_2) \cdot 1 + (-C_1 - 2C_2) \cdot 0\right] - 0 + 1 \\ &= (-2C_1 + C_2) + 1 \end{aligned}$$

So

$$(-2C_1 + C_2) + 1 = 1$$

$$-2C_1 + C_2 = 0$$

Using $C_1 = 2$:

$$-2(2) + C_2 = 0$$

$$-4 + C_2 = 0$$

$$C_2 = 4$$

Therefore the particular solution satisfying the given conditions is

$$y = e^{-2t}(2 \cos t + 4 \sin t) + \cos t + \sin t$$

3. A damped oscillator is governed by

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 6\frac{\mathrm{d}y}{\mathrm{d}t} + 13y = 75\sin 2t$$

(a) Find the general solution of this differential equation. **[6]**

(b) Show that, whatever the initial conditions, the motion is approximately of the form

$$y \approx R\sin(2t - \phi)$$

for large positive values of t, where $R > 0$ and $0 < \phi < \frac{\pi}{2}$. Determine R and ϕ. **[4]**

Solution

(a) First solve the associated homogeneous equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 6\frac{\mathrm{d}y}{\mathrm{d}t} + 13y = 0$$

Using the trial solution $y = e^{\lambda t}$ gives the auxiliary equation

$$\lambda^2 + 6\lambda + 13 = 0$$

Solving,

$$\begin{aligned}\lambda &= \frac{-6 \pm \sqrt{6^2 - 4(1)(13)}}{2} \\ &= -3 \pm 2i\end{aligned}$$

Hence the complementary function is

$$y_c = e^{-3t}(A\cos 2t + B\sin 2t)$$

For a particular integral, try

$$y_p = a\sin 2t + b\cos 2t$$

Then

$$\begin{aligned}\frac{\mathrm{d}y_p}{\mathrm{d}t} &= 2a\cos 2t - 2b\sin 2t \\ \frac{\mathrm{d}^2y_p}{\mathrm{d}t^2} &= -4a\sin 2t - 4b\cos 2t\end{aligned}$$

Substituting into the differential equation:

$$\begin{aligned}&(-4a\sin 2t - 4b\cos 2t) + 6\,(2a\cos 2t - 2b\sin 2t) + 13\,(a\sin 2t + b\cos 2t) \\ &= 75\sin 2t\end{aligned}$$

Collecting $\sin 2t$ and $\cos 2t$ terms,

$$\begin{aligned}&(-4a - 12b + 13a)\sin 2t + (12a - 4b + 13b)\cos 2t \\ &= 75\sin 2t\end{aligned}$$

So

$$(9a - 12b)\sin 2t + (12a + 9b)\cos 2t = 75\sin 2t$$

Equating coefficients gives

$$\begin{aligned} 9a - 12b &= 75 \\ 12a + 9b &= 0 \end{aligned}$$

Solve these simultaneously. From the second equation,

$$4a + 3b = 0 \quad \Rightarrow \quad b = -\frac{4a}{3}$$

Substitute into the first:

$$\begin{aligned} 9a - 12\left(-\frac{4a}{3}\right) &= 75 \\ 9a + 16a &= 75 \\ 25a &= 75 \\ a &= 3 \end{aligned}$$

Then

$$b = -\frac{4(3)}{3} = -4$$

Thus

$$y_p = 3\sin 2t - 4\cos 2t$$

Therefore the general solution is

$$y = e^{-3t}(A\cos 2t + B\sin 2t) + 3\sin 2t - 4\cos 2t$$

(b) From part (a),

$$y = e^{-3t}(A\cos 2t + B\sin 2t) + 3\sin 2t - 4\cos 2t$$

As $t \to +\infty$, the exponential factor tends to 0. Also $\sin 2t$ and $\cos 2t$ remain bounded, so

$$e^{-3t}(A\cos 2t + B\sin 2t) \longrightarrow 0$$

Therefore, whatever the initial conditions,

$$y \approx 3\sin 2t - 4\cos 2t$$

for large positive t.

Now write this in the form $R\sin(2t - \phi)$. Using the identity

$$\sin(2t - \phi) = \sin 2t\cos\phi - \cos 2t\sin\phi$$

we have

$$R\sin(2t - \phi) = R\cos\phi\ \sin 2t - R\sin\phi\ \cos 2t$$

Comparing with $3\sin 2t - 4\cos 2t$ gives

$$\begin{aligned} R\cos\phi &= 3 \\ R\sin\phi &= 4 \end{aligned}$$

Squaring and adding,

$$\begin{aligned} R^2\cos^2\phi + R^2\sin^2\phi &= 3^2 + 4^2 \\ R^2(\cos^2\phi + \sin^2\phi) &= 25 \\ R^2 &= 25 \end{aligned}$$

Since $R > 0$,

$$R = 5$$

Also,

$$\tan\phi = \frac{R\sin\phi}{R\cos\phi} = \frac{4}{3}$$

Since $0 < \phi < \frac{\pi}{2}$,

$$\phi = \arctan\left(\frac{4}{3}\right) \approx 0.927$$

Hence for large positive t,

$$y \approx 5\sin\left(2t - \arctan\frac{4}{3}\right)$$

So the required constants are $R = 5$ and $\phi = \arctan\left(\frac{4}{3}\right) \approx 0.927$.

4. A function $y(t)$ satisfies the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 2\frac{\mathrm{d}y}{\mathrm{d}t} + 5y = 10\cos t$$

Given that, when $t = 0$, $y = 5$ and $\frac{\mathrm{d}y}{\mathrm{d}t} = 2$, find the particular solution. **[10]**

Solution

First solve the associated homogeneous equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 2\frac{\mathrm{d}y}{\mathrm{d}t} + 5y = 0$$

Its auxiliary equation is

$$\lambda^2 + 2\lambda + 5 = 0$$

Solving,

$$\begin{aligned}\lambda &= \frac{-2 \pm \sqrt{2^2 - 4(1)(5)}}{2} \\ &= -1 \pm 2i\end{aligned}$$

So the complementary function is

$$y_c = e^{-t}(C_1\cos 2t + C_2 \sin 2t)$$

Now find a particular integral for

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 2\frac{\mathrm{d}y}{\mathrm{d}t} + 5y = 10\cos t$$

Since the forcing term is $10\cos t$, try

$$y_p = A\cos t + B\sin t$$

Then

$$\frac{\mathrm{d}y_p}{\mathrm{d}t} = -A\sin t + B\cos t$$

and

$$\frac{\mathrm{d}^2y_p}{\mathrm{d}t^2} = -A\cos t - B\sin t$$

Substituting into the differential equation,

$$(-A\cos t - B\sin t) + 2(-A\sin t + B\cos t) + 5(A\cos t + B\sin t) = 10\cos t$$

Collecting $\cos t$ and $\sin t$ terms,

$$\begin{aligned}(-A + 2B + 5A)\cos t + (-B - 2A + 5B)\sin t &= 10\cos t \\ (4A + 2B)\cos t + (4B - 2A)\sin t &= 10\cos t\end{aligned}$$

Equating coefficients gives

$$4A + 2B = 10, \qquad 4B - 2A = 0$$

From $4B - 2A = 0$,

$$A = 2B$$

Substitute into $4A + 2B = 10$:

$$\begin{aligned}4(2B) + 2B &= 10 \\ 8B + 2B &= 10 \\ 10B &= 10 \\ B &= 1\end{aligned}$$

Hence

$$A = 2$$

So the particular integral is

$$y_p = 2\cos t + \sin t$$

Therefore the general solution is

$$y = e^{-t}(C_1 \cos 2t + C_2 \sin 2t) + 2\cos t + \sin t$$

Now use the initial conditions.
When $t = 0$, $y = 5$:

$$\begin{aligned} 5 &= e^0(C_1 \cos 0 + C_2 \sin 0) + 2\cos 0 + \sin 0 \\ &= C_1 + 2 \end{aligned}$$

So

$$C_1 = 3$$

Next differentiate the general solution:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left[e^{-t}(C_1 \cos 2t + C_2 \sin 2t)\right] + \frac{\mathrm{d}}{\mathrm{d}t}(2\cos t + \sin t)$$

Using the product rule on the first term,

$$\begin{aligned} \frac{\mathrm{d}y}{\mathrm{d}t} &= -e^{-t}(C_1 \cos 2t + C_2 \sin 2t) + e^{-t}(-2C_1 \sin 2t + 2C_2 \cos 2t) - 2\sin t + \cos t \\ &= e^{-t}\left[(-C_1 + 2C_2)\cos 2t + (-2C_1 - C_2)\sin 2t\right] - 2\sin t + \cos t \end{aligned}$$

Now use $\frac{\mathrm{d}y}{\mathrm{d}t} = 2$ when $t = 0$:

$$\begin{aligned} 2 &= e^0\left[(-C_1 + 2C_2)\cos 0 + (-2C_1 - C_2)\sin 0\right] - 2\sin 0 + \cos 0 \\ &= (-C_1 + 2C_2) + 1 \end{aligned}$$

Substitute $C_1 = 3$:

$$\begin{aligned} 2 &= -3 + 2C_2 + 1 \\ 2 &= 2C_2 - 2 \\ 4 &= 2C_2 \\ C_2 &= 2 \end{aligned}$$

Hence the required particular solution satisfying the given conditions is

$$y(t) = e^{-t}(3\cos 2t + 2\sin 2t) + 2\cos t + \sin t$$

5. **(a)** Find the general solution of the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4\frac{\mathrm{d}y}{\mathrm{d}x} + 13y = 0 \qquad \textbf{[2]}$$

(b) Hence find the general solution of the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4\frac{\mathrm{d}y}{\mathrm{d}x} + 13y = e^{-2x}(6\cos x - 8\sin x) \qquad \textbf{[4]}$$

Solution

(a) For the homogeneous equation, use the auxiliary equation

$$\lambda^2 + 4\lambda + 13 = 0$$

Using the quadratic formula,

$$\begin{aligned}\lambda &= \frac{-4 \pm \sqrt{4^2 - 4(1)(13)}}{2} \\ &= -2 \pm 3i\end{aligned}$$

Hence the general solution is

$$y = e^{-2x}\left(A\cos 3x + B\sin 3x\right)$$

(b) From part (a), the complementary function is

$$y_c = e^{-2x}\left(C\cos 3x + D\sin 3x\right)$$

For a particular integral, try

$$y_p = e^{-2x}\left(a\cos x + b\sin x\right)$$

Then

$$\begin{aligned}\frac{\mathrm{d}y_p}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\left[e^{-2x}\left(a\cos x + b\sin x\right)\right] \\ &= -2e^{-2x}\left(a\cos x + b\sin x\right) + e^{-2x}\left(-a\sin x + b\cos x\right) \\ &= e^{-2x}\left[(b-2a)\cos x + (-a-2b)\sin x\right]\end{aligned}$$

Also,

$$\begin{aligned}\frac{\mathrm{d}^2y_p}{\mathrm{d}x^2} &= \frac{\mathrm{d}}{\mathrm{d}x}\left\{e^{-2x}\left[(b-2a)\cos x + (-a-2b)\sin x\right]\right\} \\ &= -2e^{-2x}\left[(b-2a)\cos x + (-a-2b)\sin x\right] \\ &\quad + e^{-2x}\left[(-b+2a)\sin x + (-a-2b)\cos x\right] \\ &= e^{-2x}\left[(3a-4b)\cos x + (4a+3b)\sin x\right]\end{aligned}$$

Substituting into

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4\frac{\mathrm{d}y}{\mathrm{d}x} + 13y = e^{-2x}(6\cos x - 8\sin x)$$

gives

$$\begin{aligned}&e^{-2x}\left[(3a-4b)\cos x + (4a+3b)\sin x\right] \\ &\quad + 4e^{-2x}\left[(b-2a)\cos x + (-a-2b)\sin x\right] \\ &\quad + 13e^{-2x}\left(a\cos x + b\sin x\right) \\ &= e^{-2x}(6\cos x - 8\sin x)\end{aligned}$$

So

$$e^{-2x}\left(8a\cos x + 8b\sin x\right) = e^{-2x}(6\cos x - 8\sin x)$$

Hence

$$8a\cos x + 8b\sin x = 6\cos x - 8\sin x$$

Comparing coefficients,

$$8a = 6, \qquad 8b = -8$$

So

$$a = \frac{3}{4}, \qquad b = -1$$

Therefore

$$y_p = e^{-2x}\left(\frac{3}{4}\cos x - \sin x\right)$$

Thus the general solution is

$$y = e^{-2x}\left(C\cos 3x + D\sin 3x\right) + e^{-2x}\left(\frac{3}{4}\cos x - \sin x\right)$$

So

$$y = e^{-2x}\left(C\cos 3x + D\sin 3x + \frac{3}{4}\cos x - \sin x\right)$$

6. Find the particular solution of the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 4\frac{\mathrm{d}y}{\mathrm{d}t} + 13y = 28\cos 3t - 4\sin 3t$$

given that, when $t = 0$, $y = 5$ and $\frac{\mathrm{d}y}{\mathrm{d}t} = 7$. **[10]**

Solution

We solve

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 4\frac{\mathrm{d}y}{\mathrm{d}t} + 13y = 28\cos 3t - 4\sin 3t$$

by finding the complementary function and a particular integral, then using the initial conditions. For the complementary function, solve the auxiliary equation

$$\lambda^2 + 4\lambda + 13 = 0$$

Using the quadratic formula,

$$\begin{aligned}\lambda &= \frac{-4 \pm \sqrt{4^2 - 4(1)(13)}}{2} \\ &= -2 \pm 3i\end{aligned}$$

Hence the complementary function is

$$y_c = e^{-2t}(A\cos 3t + B\sin 3t)$$

For the particular integral, since the right-hand side is of the form $28\cos 3t - 4\sin 3t$, try

$$y_p = a\cos 3t + b\sin 3t$$

Then

$$\begin{aligned}\frac{\mathrm{d}y_p}{\mathrm{d}t} &= -3a\sin 3t + 3b\cos 3t \\ \frac{\mathrm{d}^2y_p}{\mathrm{d}t^2} &= -9a\cos 3t - 9b\sin 3t\end{aligned}$$

Substituting into the differential equation,

$$\begin{aligned}\frac{\mathrm{d}^2y_p}{\mathrm{d}t^2} + 4\frac{\mathrm{d}y_p}{\mathrm{d}t} + 13y_p &= (-9a\cos 3t - 9b\sin 3t) + 4(-3a\sin 3t + 3b\cos 3t) + 13(a\cos 3t + b\sin 3t) \\ &= (-9a + 12b + 13a)\cos 3t + (-9b - 12a + 13b)\sin 3t \\ &= (4a + 12b)\cos 3t + (4b - 12a)\sin 3t\end{aligned}$$

Equating coefficients with $28\cos 3t - 4\sin 3t$ gives

$$\begin{aligned}4a + 12b &= 28 \\ 4b - 12a &= -4\end{aligned}$$

Dividing both equations by 4,

$$\begin{aligned}a + 3b &= 7 \\ b - 3a &= -1\end{aligned}$$

From the second equation,

$$b = 3a - 1$$

Substitute into the first:

$$\begin{aligned} a + 3(3a - 1) &= 7 \\ a + 9a - 3 &= 7 \\ 10a &= 10 \\ a &= 1 \end{aligned}$$

Then

$$b = 3(1) - 1 = 2$$

So

$$y_p = \cos 3t + 2\sin 3t$$

Hence the general solution is

$$y = e^{-2t}(A\cos 3t + B\sin 3t) + \cos 3t + 2\sin 3t$$

Now use the initial conditions.
First, when $t = 0$, $y = 5$:

$$\begin{aligned} 5 &= e^0(A\cos 0 + B\sin 0) + \cos 0 + 2\sin 0 \\ &= A(1) + B(0) + 1 + 0 \\ &= A + 1 \end{aligned}$$

So

$$A = 4$$

Next differentiate y.
For the homogeneous part,

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}\left[e^{-2t}(A\cos 3t + B\sin 3t)\right] &= e^{-2t}\,(-2)\,(A\cos 3t + B\sin 3t) + e^{-2t}\,(-3A\sin 3t + 3B\cos 3t) \\ &= e^{-2t}\left[(3B - 2A)\cos 3t + (-3A - 2B)\sin 3t\right] \end{aligned}$$

Also,

$$\frac{\mathrm{d}}{\mathrm{d}t}\,(\cos 3t + 2\sin 3t) = -3\sin 3t + 6\cos 3t$$

Therefore

$$\frac{\mathrm{d}y}{\mathrm{d}t} = e^{-2t}\left[(3B - 2A)\cos 3t + (-3A - 2B)\sin 3t\right] - 3\sin 3t + 6\cos 3t$$

Using $\frac{\mathrm{d}y}{\mathrm{d}t} = 7$ when $t = 0$,

$$\begin{aligned} 7 &= e^0\left[(3B - 2A)\cos 0 + (-3A - 2B)\sin 0\right] - 3\sin 0 + 6\cos 0 \\ &= (3B - 2A)(1) + 0 + 0 + 6 \\ &= 3B - 2A + 6 \end{aligned}$$

Substitute $A = 4$:

$$\begin{aligned} 7 &= 3B - 8 + 6 \\ 7 &= 3B - 2 \\ 3B &= 9 \\ B &= 3 \end{aligned}$$

Hence the required particular solution is

$$y = (4\cos 3t + 3\sin 3t)e^{-2t} + \cos 3t + 2\sin 3t$$

7. Solve the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{\mathrm{d}y}{\mathrm{d}x} - 6y = 50\sin x$$

subject to the conditions $y(0) = 3$ and $y'(0) = -6$. **[10]**

Solution

We solve

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{\mathrm{d}y}{\mathrm{d}x} - 6y = 50\sin x$$

subject to

$$y(0) = 3, \qquad y'(0) = -6$$

First, solve the homogeneous equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{\mathrm{d}y}{\mathrm{d}x} - 6y = 0$$

Its auxiliary equation is

$$\lambda^2 - \lambda - 6 = 0$$

Factorising,

$$(\lambda - 3)(\lambda + 2) = 0$$

So the roots are $\lambda = 3$ and $\lambda = -2$. Hence the complementary function is

$$y_c = Ae^{3x} + Be^{-2x}$$

Now find a particular integral. Since the right-hand side is $50\sin x$, try

$$y_p = a\sin x + b\cos x$$

Then

$$\frac{\mathrm{d}y_p}{\mathrm{d}x} = a\cos x - b\sin x$$
$$\frac{\mathrm{d}^2y_p}{\mathrm{d}x^2} = -a\sin x - b\cos x$$

Substituting into the differential equation,

$$\begin{aligned}\frac{\mathrm{d}^2y_p}{\mathrm{d}x^2} - \frac{\mathrm{d}y_p}{\mathrm{d}x} - 6y_p &= (-a\sin x - b\cos x) - (a\cos x - b\sin x) - 6\,(a\sin x + b\cos x) \\ &= -a\sin x - b\cos x - a\cos x + b\sin x - 6a\sin x - 6b\cos x \\ &= (b - 7a)\sin x + (-a - 7b)\cos x\end{aligned}$$

We want this to equal $50\sin x$, so equating coefficients gives

$$\begin{aligned} b - 7a &= 50 \\ -a - 7b &= 0\end{aligned}$$

From the second equation,

$$a = -7b$$

Substitute into the first:

$$\begin{aligned} b - 7(-7b) &= 50 \\ b + 49b &= 50 \\ 50b &= 50 \\ b &= 1\end{aligned}$$

So

$$a = -7$$

Therefore the particular integral is

$$y_p = -7\sin x + \cos x$$

Hence the general solution is

$$y = Ae^{3x} + Be^{-2x} - 7\sin x + \cos x$$

Now use the initial conditions. First,

$$\begin{aligned} y(0) &= Ae^0 + Be^0 - 7\sin 0 + \cos 0 \\ &= A + B + 1 \end{aligned}$$

Since $y(0) = 3$,

$$A + B + 1 = 3$$

so

$$A + B = 2$$

Next differentiate the general solution:

$$y'(x) = 3Ae^{3x} - 2Be^{-2x} - 7\cos x - \sin x$$

Now apply $y'(0) = -6$:

$$\begin{aligned} y'(0) &= 3Ae^0 - 2Be^0 - 7\cos 0 - \sin 0 \\ &= 3A - 2B - 7 \end{aligned}$$

So

$$3A - 2B - 7 = -6$$

hence

$$3A - 2B = 1$$

We now solve

$$\begin{aligned} A + B &= 2 \\ 3A - 2B &= 1 \end{aligned}$$

Multiply the first equation by 2:

$$2A + 2B = 4$$

Add this to $3A - 2B = 1$:

$$\begin{aligned} (2A + 2B) + (3A - 2B) &= 4 + 1 \\ 5A &= 5 \\ A &= 1 \end{aligned}$$

Then from $A + B = 2$,

$$\begin{aligned} 1 + B &= 2 \\ B &= 1 \end{aligned}$$

Therefore the required solution is

$$y = e^{3x} + e^{-2x} - 7\sin x + \cos x$$

8. Find the particular solution of the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 4\frac{\mathrm{d}y}{\mathrm{d}t} + 13y = 8e^{-2t}\sin t$$

given that, when $t = 0$, $y = 1$ and $\frac{\mathrm{d}y}{\mathrm{d}t} = 2$. **[10]**

Solution

For the homogeneous part,

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 4\frac{\mathrm{d}y}{\mathrm{d}t} + 13y = 0$$

The auxiliary equation is

$$\lambda^2 + 4\lambda + 13 = 0$$

So

$$\lambda = \frac{-4 \pm \sqrt{16 - 52}}{2} = -2 \pm 3i$$

Hence

$$y_c = e^{-2t}\left(A\cos 3t + B\sin 3t\right)$$

For a particular integral, try

$$y_p = e^{-2t}\left(a\sin t + b\cos t\right)$$

Then

$$\begin{aligned}\frac{\mathrm{d}y_p}{\mathrm{d}t} &= \frac{\mathrm{d}}{\mathrm{d}t}\left[e^{-2t}\left(a\sin t + b\cos t\right)\right] \\ &= -2e^{-2t}\left(a\sin t + b\cos t\right) + e^{-2t}\left(a\cos t - b\sin t\right) \\ &= e^{-2t}\left[(-b - 2a)\sin t + (a - 2b)\cos t\right]\end{aligned}$$

Also,

$$\begin{aligned}\frac{\mathrm{d}^2y_p}{\mathrm{d}t^2} &= \frac{\mathrm{d}}{\mathrm{d}t}\left\{e^{-2t}\left[(-b - 2a)\sin t + (a - 2b)\cos t\right]\right\} \\ &= -2e^{-2t}\left[(-b - 2a)\sin t + (a - 2b)\cos t\right] \\ &\quad + e^{-2t}\left[(-b - 2a)\cos t - (a - 2b)\sin t\right] \\ &= e^{-2t}\left[(3a + 4b)\sin t + (-4a + 3b)\cos t\right]\end{aligned}$$

Substituting into

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + 4\frac{\mathrm{d}y}{\mathrm{d}t} + 13y = 8e^{-2t}\sin t$$

gives

$$\begin{aligned}&e^{-2t}\left[(3a + 4b)\sin t + (-4a + 3b)\cos t\right] \\ &\quad + 4e^{-2t}\left[(-b - 2a)\sin t + (a - 2b)\cos t\right] \\ &\quad + 13e^{-2t}\left(a\sin t + b\cos t\right) \\ &= 8e^{-2t}\sin t\end{aligned}$$

So

$$e^{-2t}\left(8a\sin t + 8b\cos t\right) = 8e^{-2t}\sin t$$

Hence

$$8a\sin t + 8b\cos t = 8\sin t$$

Comparing coefficients,

$$8a = 8, \qquad 8b = 0$$

So

$$a = 1, \qquad b = 0$$

Therefore

$$y_p = e^{-2t} \sin t$$

Thus the general solution is

$$y = e^{-2t} (A \cos 3t + B \sin 3t) + e^{-2t} \sin t$$

So

$$y = e^{-2t} (A \cos 3t + B \sin 3t + \sin t)$$

Now use the initial conditions.
Given $y(0) = 1$,

$$1 = e^{0} (A \cos 0 + B \sin 0 + \sin 0)$$

So

$$1 = A$$

Hence

$$A = 1$$

Now differentiate:

$$\begin{aligned} \frac{\mathrm{d}y}{\mathrm{d}t} &= \frac{\mathrm{d}}{\mathrm{d}t} \left[e^{-2t} (A \cos 3t + B \sin 3t + \sin t) \right] \\ &= -2e^{-2t} (A \cos 3t + B \sin 3t + \sin t) \\ &\quad + e^{-2t} (-3A \sin 3t + 3B \cos 3t + \cos t) \\ &= e^{-2t} (-3A \sin 3t + 3B \cos 3t + \cos t - 2A \cos 3t - 2B \sin 3t - 2 \sin t) \end{aligned}$$

At $t = 0$,

$$\frac{\mathrm{d}y}{\mathrm{d}t}(0) = 3B + 1 - 2A$$

Since $\frac{\mathrm{d}y}{\mathrm{d}t}(0) = 2$ and $A = 1$,

$$2 = 3B + 1 - 2$$

So

$$2 = 3B - 1$$

Hence

$$3B = 3$$

So

$$B = 1$$

Substituting $A = 1$ and $B = 1$, the required particular solution is

$$y = e^{-2t} \big(\cos 3t + \sin 3t + \sin t \big)$$

9. **(a)** Show that, for all real x,

$$(1+\tanh x)\cosh x = e^x$$ **[2]**

(b) Hence solve the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 2\frac{\mathrm{d}y}{\mathrm{d}x} + y = 3(1+\tanh x)\cosh x$$

given that, when $x = 0$, $y = 1$ and $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$. **[10]**

Solution

(a) Using $\tanh x = \dfrac{\sinh x}{\cosh x}$,

$$\begin{aligned}(1+\tanh x)\cosh x &= \left(1+\frac{\sinh x}{\cosh x}\right)\cosh x \\ &= \cosh x + \sinh x \\ &= \frac{e^x+e^{-x}}{2} + \frac{e^x-e^{-x}}{2} \\ &= e^x\end{aligned}$$

Hence, for all real x,

$$(1+\tanh x)\cosh x = e^x$$

(b) From part (a), the differential equation becomes

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 2\frac{\mathrm{d}y}{\mathrm{d}x} + y = 3e^x$$

For the homogeneous equation,

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 2\frac{\mathrm{d}y}{\mathrm{d}x} + y = 0$$

The auxiliary equation is

$$\lambda^2 - 2\lambda + 1 = 0$$

So

$$(\lambda-1)^2 = 0$$

Hence the complementary function is

$$y_c = (A+Bx)e^x$$

For a particular integral, since the right-hand side is $3e^x$ and $\lambda = 1$ is a repeated root, try

$$y_p = Cx^2e^x$$

Then

$$\begin{aligned}\frac{\mathrm{d}y_p}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\left(Cx^2e^x\right) \\ &= 2Cxe^x + Cx^2e^x \\ &= e^x\left(Cx^2+2Cx\right)\end{aligned}$$

Differentiate again:

$$\begin{aligned}\frac{\mathrm{d}^2y_p}{\mathrm{d}x^2} &= \frac{\mathrm{d}}{\mathrm{d}x}\left[e^x\left(Cx^2+2Cx\right)\right]\\ &= e^x\left(Cx^2+2Cx\right)+e^x\left(2Cx+2C\right)\\ &= e^x\left(Cx^2+4Cx+2C\right)\end{aligned}$$

Substituting into the differential equation gives

$$\begin{aligned}\frac{\mathrm{d}^2y_p}{\mathrm{d}x^2}-2\frac{\mathrm{d}y_p}{\mathrm{d}x}+y_p &= e^x\left(Cx^2+4Cx+2C\right)-2e^x\left(Cx^2+2Cx\right)+Cx^2e^x\\ &= 2Ce^x\end{aligned}$$

So

$$2Ce^x = 3e^x$$

Hence

$$2C = 3$$

So

$$C = \frac{3}{2}$$

Therefore

$$y_p = \frac{3x^2}{2}e^x$$

Thus the general solution is

$$y = (A+Bx)e^x + \frac{3x^2}{2}e^x$$

So

$$y = e^x\left(A+Bx+\frac{3x^2}{2}\right)$$

Now use the initial conditions.
From $y(0)=1$,

$$\begin{aligned}1 &= e^0\left(A+B\cdot 0+\frac{3(0)^2}{2}\right)\\ &= A\end{aligned}$$

So

$$A = 1$$

Now differentiate:

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\left[e^x\left(A+Bx+\frac{3x^2}{2}\right)\right]\\ &= e^x\left(A+Bx+\frac{3x^2}{2}\right)+e^x(B+3x)\\ &= e^x\left(A+Bx+\frac{3x^2}{2}+B+3x\right)\end{aligned}$$

At $x=0$,

$$\frac{\mathrm{d}y}{\mathrm{d}x}(0) = A+B$$

Since $\frac{\mathrm{d}y}{\mathrm{d}x}(0)=0$ and $A=1$,

$$0 = 1+B$$

Hence

$$B = -1$$

Substituting $A=1$ and $B=-1$, the required solution is

$$y = e^x\left(\frac{3x^2}{2}-x+1\right)$$

10. **(a)** Show that

$$\frac{\cosh x - \sinh x}{\cosh x + \sinh x} = e^{-2x}$$

[2]

(b) Hence find the particular solution of the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 4\left(\frac{\cosh x - \sinh x}{\cosh x + \sinh x}\right)$$

given that, when $x = 0$, $y = 1$ and $\frac{\mathrm{d}y}{\mathrm{d}x} = 1$. **[10]**

Solution

(a) Using

$$\cosh x = \frac{e^x + e^{-x}}{2} \qquad \text{and} \qquad \sinh x = \frac{e^x - e^{-x}}{2}$$

we get

$$\begin{aligned}\cosh x - \sinh x &= \frac{e^x + e^{-x}}{2} - \frac{e^x - e^{-x}}{2} \\ &= \frac{e^x + e^{-x} - e^x + e^{-x}}{2} \\ &= \frac{2e^{-x}}{2} \\ &= e^{-x}\end{aligned}$$

Also,

$$\begin{aligned}\cosh x + \sinh x &= \frac{e^x + e^{-x}}{2} + \frac{e^x - e^{-x}}{2} \\ &= \frac{e^x + e^{-x} + e^x - e^{-x}}{2} \\ &= \frac{2e^x}{2} \\ &= e^x\end{aligned}$$

Therefore,

$$\frac{\cosh x - \sinh x}{\cosh x + \sinh x} = \frac{e^{-x}}{e^x} = e^{-2x}$$

Hence,

$$\frac{\cosh x - \sinh x}{\cosh x + \sinh x} = e^{-2x}$$

(b) From part (a), the differential equation becomes

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 4e^{-2x}$$

For the homogeneous equation,

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 4\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 0$$

The auxiliary equation is

$$\lambda^2 + 4\lambda + 4 = 0$$

So

$$(\lambda + 2)^2 = 0$$

Hence the complementary function is

$$y_c = (Cx + D)e^{-2x}$$

Since the right-hand side is $4e^{-2x}$ and -2 is a repeated root of the auxiliary equation, try

$$y_p = Ax^2e^{-2x}$$

Differentiate:

$$\begin{aligned}\frac{\mathrm{d}y_p}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\left(Ax^2e^{-2x}\right)\\ &= 2Axe^{-2x} - 2Ax^2e^{-2x}\\ &= e^{-2x}(2Ax - 2Ax^2)\end{aligned}$$

Differentiate again:

$$\begin{aligned}\frac{\mathrm{d}^2y_p}{\mathrm{d}x^2} &= \frac{\mathrm{d}}{\mathrm{d}x}\left[e^{-2x}(2Ax - 2Ax^2)\right]\\ &= -2e^{-2x}(2Ax - 2Ax^2) + e^{-2x}(2A - 4Ax)\\ &= e^{-2x}(2A - 8Ax + 4Ax^2)\end{aligned}$$

Substituting into the differential equation gives

$$\begin{aligned}\frac{\mathrm{d}^2y_p}{\mathrm{d}x^2} + 4\frac{\mathrm{d}y_p}{\mathrm{d}x} + 4y_p &= e^{-2x}(2A - 8Ax + 4Ax^2) + 4e^{-2x}(2Ax - 2Ax^2) + 4Ax^2e^{-2x}\\ &= e^{-2x}(2A)\end{aligned}$$

So

$$2Ae^{-2x} = 4e^{-2x}$$

Hence

$$2A = 4$$

So

$$A = 2$$

Therefore

$$y_p = 2x^2e^{-2x}$$

Thus the general solution is

$$y = (Cx + D)e^{-2x} + 2x^2e^{-2x}$$

So

$$y = e^{-2x}(2x^2 + Cx + D)$$

Now use the initial conditions.
Since $y(0) = 1$,

$$\begin{aligned}1 &= e^0(2 \cdot 0^2 + C \cdot 0 + D)\\ &= D\end{aligned}$$

So

$$D = 1$$

Now differentiate:

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\left[e^{-2x}(2x^2 + Cx + D)\right]\\ &= -2e^{-2x}(2x^2 + Cx + D) + e^{-2x}(4x + C)\\ &= e^{-2x}(4x + C - 4x^2 - 2Cx - 2D)\end{aligned}$$

At $x = 0$,

$$\frac{\mathrm{d}y}{\mathrm{d}x}(0) = C - 2D$$

Since $\frac{\mathrm{d}y}{\mathrm{d}x}(0) = 1$ and $D = 1$,

$$1 = C - 2$$

Hence

$$C = 3$$

Substituting $C = 3$ and $D = 1$, the required particular solution is

$$y = (2x^2 + 3x + 1)e^{-2x}$$

11. **(a)** Show that

$$3\cosh x - \sinh x = e^x + 2e^{-x}$$

[2]

(b) Hence solve the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{\mathrm{d}y}{\mathrm{d}x} - 2y = 6(3\cosh x - \sinh x)$$

given that, when $x = 0$, $y = 1$ and $\frac{\mathrm{d}y}{\mathrm{d}x} = 1$. **[10]**

Solution

(a) Using

$$\cosh x = \frac{e^x + e^{-x}}{2} \qquad \text{and} \qquad \sinh x = \frac{e^x - e^{-x}}{2}$$

we have

$$\begin{aligned} 3\cosh x - \sinh x &= 3\left(\frac{e^x + e^{-x}}{2}\right) - \left(\frac{e^x - e^{-x}}{2}\right) \\ &= \frac{3e^x + 3e^{-x} - e^x + e^{-x}}{2} \\ &= \frac{2e^x + 4e^{-x}}{2} \\ &= e^x + 2e^{-x} \end{aligned}$$

Hence

$$3\cosh x - \sinh x = e^x + 2e^{-x}$$

(b) From part (a),

$$6(3\cosh x - \sinh x) = 6(e^x + 2e^{-x}) = 6e^x + 12e^{-x}$$

so the differential equation becomes

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{\mathrm{d}y}{\mathrm{d}x} - 2y = 6e^x + 12e^{-x}$$

First solve the homogeneous equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{\mathrm{d}y}{\mathrm{d}x} - 2y = 0$$

The auxiliary equation is

$$\lambda^2 - \lambda - 2 = 0$$

Factorising,

$$\begin{aligned} \lambda^2 - \lambda - 2 &= 0 \\ (\lambda - 2)(\lambda + 1) &= 0 \end{aligned}$$

So the roots are $\lambda = 2$ and $\lambda = -1$. Therefore the complementary function is

$$y_c = Ae^{2x} + Be^{-x}$$

Now find a particular integral. Since the right-hand side is $6e^x + 12e^{-x}$, take

$$y_p = Ce^x + Dxe^{-x}$$

We use Dxe^{-x} rather than De^{-x} because e^{-x} is already part of the complementary function.

For the Ce^x term,

$$\frac{\mathrm{d}}{\mathrm{d}x}(Ce^x) = Ce^x, \qquad \frac{\mathrm{d}^2}{\mathrm{d}x^2}(Ce^x) = Ce^x$$

Substituting into the left-hand side,

$$\begin{aligned}\frac{\mathrm{d}^2}{\mathrm{d}x^2}(Ce^x) - \frac{\mathrm{d}}{\mathrm{d}x}(Ce^x) - 2(Ce^x) &= Ce^x - Ce^x - 2Ce^x \\ &= -2Ce^x\end{aligned}$$

This must equal $6e^x$, so

$$-2C = 6$$

Hence

$$C = -3$$

For the Dxe^{-x} term,

$$\begin{aligned}y &= Dxe^{-x} \\ \frac{\mathrm{d}y}{\mathrm{d}x} &= D\frac{\mathrm{d}}{\mathrm{d}x}(xe^{-x}) \\ &= D\left(e^{-x} - xe^{-x}\right) \\ &= D(1-x)e^{-x}\end{aligned}$$

Differentiate again:

$$\begin{aligned}\frac{\mathrm{d}^2y}{\mathrm{d}x^2} &= D\frac{\mathrm{d}}{\mathrm{d}x}\left((1-x)e^{-x}\right) \\ &= D\left((-1)e^{-x} + (1-x)(-e^{-x})\right) \\ &= D(x-2)e^{-x}\end{aligned}$$

Substituting into the left-hand side,

$$\begin{aligned}\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - \frac{\mathrm{d}y}{\mathrm{d}x} - 2y &= D(x-2)e^{-x} - D(1-x)e^{-x} - 2Dxe^{-x} \\ &= D\big(x - 2 - 1 + x - 2x\big)e^{-x} \\ &= -3De^{-x}\end{aligned}$$

This must equal $12e^{-x}$, so

$$-3D = 12$$

Hence

$$D = -4$$

Therefore a particular integral is

$$y_p = -3e^x - 4xe^{-x}$$

and the general solution is

$$y = Ae^{2x} + Be^{-x} - 3e^x - 4xe^{-x}$$

Now use the initial conditions.

When $x = 0$, $y = 1$:

$$\begin{aligned}1 &= Ae^0 + Be^0 - 3e^0 - 4(0)e^0 \\ &= A + B - 3\end{aligned}$$

So

$$A + B = 4$$

Differentiate the general solution:

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} &= 2Ae^{2x} - Be^{-x} - 3e^x - 4\frac{\mathrm{d}}{\mathrm{d}x}(xe^{-x}) \\ &= 2Ae^{2x} - Be^{-x} - 3e^x - 4(1-x)e^{-x}\end{aligned}$$

When $x = 0$, $\frac{\mathrm{d}y}{\mathrm{d}x} = 1$:

$$\begin{aligned} 1 &= 2Ae^0 - Be^0 - 3e^0 - 4(1-0)e^0 \\ &= 2A - B - 3 - 4 \\ &= 2A - B - 7 \end{aligned}$$

So

$$2A - B = 8$$

Solve the simultaneous equations

$$\begin{aligned} A + B &= 4 \\ 2A - B &= 8 \end{aligned}$$

Adding gives

$$3A = 12$$

so

$$A = 4$$

Then

$$B = 4 - A = 0$$

Substituting back, the required solution is

$$y = 4e^{2x} - 3e^x - 4xe^{-x}$$

12. Determine the solution $y(x)$ of the initial value problem

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 4\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = xe^{2x}$$

subject to

$$y(0) = 1, \qquad \left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=0} = 0$$

[10]

Solution

For the homogeneous equation,

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 4\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = 0$$

The auxiliary equation is

$$\lambda^2 - 4\lambda + 4 = 0$$

So

$$(\lambda - 2)^2 = 0$$

Hence the complementary function is

$$y_c = (A + Bx)e^{2x}$$

For a particular integral, since the right-hand side is xe^{2x} and $\lambda = 2$ is a repeated root, try

$$y_p = e^{2x}(ax^3 + bx^2)$$

Then

$$\begin{aligned}\frac{\mathrm{d}y_p}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\left[e^{2x}(ax^3 + bx^2)\right] \\ &= 2e^{2x}(ax^3 + bx^2) + e^{2x}(3ax^2 + 2bx) \\ &= e^{2x}\left(2ax^3 + (2b + 3a)x^2 + 2bx\right)\end{aligned}$$

Differentiate again:

$$\begin{aligned}\frac{\mathrm{d}^2y_p}{\mathrm{d}x^2} &= \frac{\mathrm{d}}{\mathrm{d}x}\left[e^{2x}\left(2ax^3 + (2b + 3a)x^2 + 2bx\right)\right] \\ &= 2e^{2x}\left(2ax^3 + (2b + 3a)x^2 + 2bx\right) \\ &\quad + e^{2x}\left(6ax^2 + 2(2b + 3a)x + 2b\right) \\ &= e^{2x}\left(4ax^3 + (4b + 12a)x^2 + (8b + 6a)x + 2b\right)\end{aligned}$$

Substituting into the differential equation gives

$$\begin{aligned}\frac{\mathrm{d}^2y_p}{\mathrm{d}x^2} - 4\frac{\mathrm{d}y_p}{\mathrm{d}x} + 4y_p &= e^{2x}\left(4ax^3 + (4b + 12a)x^2 + (8b + 6a)x + 2b\right) \\ &\quad - 4e^{2x}\left(2ax^3 + (2b + 3a)x^2 + 2bx\right) \\ &\quad + 4e^{2x}(ax^3 + bx^2)\end{aligned}$$

So

$$\frac{\mathrm{d}^2y_p}{\mathrm{d}x^2} - 4\frac{\mathrm{d}y_p}{\mathrm{d}x} + 4y_p = e^{2x}(6ax + 2b)$$

Comparing with xe^{2x}, we get

$$6a = 1, \qquad 2b = 0$$

So

$$a = \frac{1}{6}, \qquad b = 0$$

Therefore

$$y_p = \frac{x^3}{6}e^{2x}$$

Thus the general solution is

$$y = (A + Bx)e^{2x} + \frac{x^3}{6}e^{2x}$$

So

$$y = e^{2x}\left(A + Bx + \frac{x^3}{6}\right)$$

Now use the initial conditions.
From $y(0) = 1$,

$$\begin{aligned} 1 &= e^0\left(A + B \cdot 0 + \frac{0^3}{6}\right) \\ &= A \end{aligned}$$

So

$$A = 1$$

Now differentiate:

$$\begin{aligned} \frac{\mathrm{d}y}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\left[e^{2x}\left(A + Bx + \frac{x^3}{6}\right)\right] \\ &= 2e^{2x}\left(A + Bx + \frac{x^3}{6}\right) + e^{2x}\left(B + \frac{x^2}{2}\right) \\ &= e^{2x}\left(2A + 2Bx + \frac{x^3}{3} + B + \frac{x^2}{2}\right) \end{aligned}$$

At $x = 0$,

$$\frac{\mathrm{d}y}{\mathrm{d}x}(0) = 2A + B$$

Since $\frac{\mathrm{d}y}{\mathrm{d}x}(0) = 0$ and $A = 1$,

$$0 = 2 + B$$

Hence

$$B = -2$$

Substituting $A = 1$ and $B = -2$, the solution of the initial value problem is

$$y(x) = e^{2x}\left(\frac{x^3}{6} - 2x + 1\right)$$